

Enterprise Pro – Coursework: Part A

Requirements, specifications and prototype implementation

Hand out: **Monday 9th February**

Hand in: **Thursday, 12th March @ 5pm**

Assignment weight is 50% of the overall mark.

Assignment Description:

I. Projects

Each team will work for a project with a client (industry sponsor). Based on the preferences expressed by the teams and the brief justification evaluated by the industry sponsors, the projects are allocated to the teams.

The clients are interviewed by the teams, and the Requirements and Specifications documents will be produced by the above mentioned deadline. The iterative process of producing the software products will include two (interim and final) delivery stages when the clients and supervisor(s) will provide feedback to each team. Staff members (supervisors) and industry sponsors will assess regularly the progress made by the teams providing feedback. The final product and the associated documentation will be finally delivered and demonstrated in the last week (12) of the semester.

II. Activities

Each team will produce and deliver project documentation – requirements document; project plan (e.g., Gantt chart), generate meeting minutes with the tasks and responsibilities; and a software system reflecting the requirements and the client's needs. In each case, there is a group activity component and an individual one as well. The marks of individuals might be different in the same group when different measurable outputs have been produced.

III. Deliverables

Based on the brief description of the systems and the additional information obtained during the interview session each team is expected to submit, by the above mentioned deadline, the following:

A. Requirements document: This document must be submitted via **Canvas**.

A.1. Requirements Specifications and (Interim) Prototype Implementation documentation following the structure discussed in lectures and lab notes (weeks 3 and 4): It will include the following: introduction, team expertise, the rationale (reasons) of topic choice, literature review (e.g. similar existing solutions, tools, equipment), functional (and non-functional) requirements, data description, interface, work plan, and reflections on legal, social, ethical and professional issues (LSEPI) and other risk and economic aspects for your project.

A.2. Peer-review. You should include a **peer-review** summarising the contribution of each team member by describing the **tasks allocated**, **achievements**, and **scoring the contribution** (with a **grade** in the **range 1..10**, where 1 means minimal contribution and 10 an outstanding one). If a team member does not contribute to the work, a mark of 0 will be given. If the contribution is not partial, then a partial mark will be given.

B. The code associated with the interim prototype system resulted from the first iteration: accessible via **GitHub** (under the **Software** directory of your GitHub folder). The prototype and code progress to date will be demonstrated to the academic supervisor / the industry sponsor in week 8.

C. Weekly meeting minutes of team's activity (presenting task allocations, and progress on tasks both team-level and individual), NDA, and an optional video recording of the prototype functionalities (implemented to date) demonstration (as an optional back-up for the prototype demonstration to the industry sponsor): These must be stored in the **Minutes**, **NDA**, **Videos** folders of your **Github** repository, respectively. **Please just provide the links in the report, do not submit them as part of the report.**

IV. Remarks.

The following will be made available by the date indicated above:

1. Names of the deliverables (submitted documents) will be of the form: RequirementsTx (where x is your Team number).
2. The documentation will contain a cover page with team's id (name, number, and flag); members list (full name, UoB#, UoB email address) and project choice.
3. All requested assignment documents must be submitted by only **ONE** team member on Canvas Assignments section for marking and feedback.

V. Marking scheme.

A mark out of 100% for CW1 will be made of marks for components A-C with the following weights: $CW1 = A + B + C$

A. Requirements Specifications and (Interim) Prototype Implementation (50 marks):

1.1 The following marks will be assigned on quality and meaningfulness for each of:

- project brief **(2.5 marks)**
- introduction of team expertise and rationale of topic choice **(2.5 marks)**
- work plan **(2.5 marks)**
- peer review **(2.5 marks)**

1.2 Adequate structure and layout (title; authors; chapters on functions, use cases, data, interface, non-functional aspects/constraints; formatting used consistently): **(5 marks)**

1.3 Coherent description of the requirements, specifications, functions, non-functional requirements or any constraints captured so far: **(10 marks)**

1.4 Coherent description of the data **(10 marks)**

1.5 Coherent description of interfaces design in correlation with functions **(5 marks)**

1.6 Reflection on legal, social, ethical, security, professional issues of the project, risk and economic aspects, environment awareness **(10 marks)**

B. Interim Prototype (35 marks)

2.1 Significant functionality implemented – either front-end or back-end, or a combination of both; fully functional **(30 marks)**

2.2 Code well-structured, with comments, OO where is the case; clear separation back-end and front-end **(5 marks)**

C. Team Activity: NDAs, and meeting minutes (15 marks)

3.1 Adequate structure and style of the minutes (as presented in lectures) including task allocation and systematic meeting notes: **(12.5 marks)**

3.2. NDA **(2.5 marks)**

Remark: The following grading will be used (based on 10-scale):

[0-3 marks]: [no/lack of] relevant content

[4-5 marks]: satisfactory content lacking consistent details though

[6-7 marks]: good consistent content and level of detail

[8-10 marks]: significant level of consistency and high level quality of detail

Enterprise Pro Learning Outcomes and how these will be achieved:

LO1. Demonstrate a sound theoretical understanding of software project development through teamwork and the practical ability to implement, document and test a relatively complex software product

LO2. Contribute to the design and development of a good quality software application by using appropriate methods and tools, through teamwork.

LO3. Work collaboratively to solve problems and develop adequate communication skills.

LO4. Critically analyse professional, legal and ethical issues.

The students are working in teams (as mentioned in all LOs), demonstrating 'sound theoretical understanding of software project development '(LO1) by applying software engineering methods to the project development life cycle (illustrated by the requirements and testing documents, software coding). Reflection on LSEPI and other risk and economic aspects (LO4) is present in the requirements document. The quality of the software application (LO2) is checked on a regular basis through supervision meetings and assessed by clients and staff members, at the end of each of the two stages. The collaborative work (LO3) is consistently described by team meeting minutes, where project planning and task allocation are also presented. All these achievements are assessed, as reflected by marking schemes.